

VARSHA KALEESWARAN

B.Tech – Artificial Intelligence & Data Science | CGPA: 9.17 | Coimbatore, TN

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PROFESSIONAL SUMMARY

Motivated final-year B.Tech AI & Data Science student with multiple internships and proven ML/DL expertise. Built production RAG pipeline with 1000+ vectors and real-time memory system; engineered SaaS AI mentorship platform using Groq API & Llama 3.3 70B; optimized edge AI models achieving 39% size reduction & 50% FPS improvement. Pentathlon 2025 Finalist (AICTE) | SIH 2025 Internally Selected | Flipkart GRID 7.0 | Skilled in Python, TensorFlow, PyTorch, LangChain, FastAPI, Streamlit, and production deployment.

TECHNICAL SKILLS

Programming: Python, Java & C(Basics) , SQL

AI/ML & GenAI: Machine Learning, Deep Learning (CNN, RNN), Generative AI, LLMs, RAG, Prompt Engineering, NLP, Computer Vision, EDA, Feature Engineering, Data Visualization

Frameworks & Libraries: LangChain, ChromaDB, TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, FastAPI, Streamlit

Web & Databases: HTML, CSS, JavaScript, ReactJS, MySQL, DBMS (Basics of all), **Core CS:** Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Software Engineering

Cloud & Tools: AWS, Git, GitHub, REST APIs, Postman, VS Code, Jupyter Notebook, Google Colab, Arduino IDE

Others: Problem Solving, Analytical Thinking, Teamwork, Communication, Leadership

PROFESSIONAL EXPERIENCE

AI & Robotics Intern | iHub Robotics | [Certificate](#)

May 2025 – Jun 2025 | Coimbatore, Tamil Nadu

- Developed **VisionBot**, an AI-powered humanoid robotic head with real-time face detection and object tracking using **OpenCV** and **Arduino**, enabling autonomous head and eye movement for enhanced human-robot interaction.
- Engineered a **dual conveyor belt system with a robotic arm** for automated object handling by integrating embedded systems, optimizing hardware calibration, debugging, and real-time control.

AI Intern | CodeAlpha | [Certificate](#)

Dec 2025 – Jan 2026

- Developed end-to-end **Machine Learning** and **Natural Language Processing (NLP)** applications, including **Sentiment Analysis** and **Salary Prediction**, by implementing data preprocessing, feature engineering, model training, evaluation, and predictive modeling.
- Optimized model performance through feature selection, hyperparameter tuning, comparative analysis, debugging, and technical documentation while following software development best practices.

AI/ML Intern | Pirai Infotech Pvt. Ltd. | [Certificate](#)

Mar 2026 | Coimbatore, Tamil Nadu

- Developed a **Retrieval-Augmented Generation (RAG)** application using **LangChain** and **ChromaDB**, implementing vector embeddings, semantic search, and document chunking to enable accurate, context-aware question answering while reducing LLM hallucinations.
- Built an end-to-end AI retrieval pipeline by integrating **Large Language Models (LLMs)**, prompt engineering, **FastAPI**, and **Streamlit**, optimizing retrieval relevance, response quality, and application scalability through iterative testing and performance tuning.

KEY PROJECTS.

RAG Chatbot & AI Memory Assistant

[Github](#)

- Identified gap: standard AI chatbots lack persistent memory, forgetting conversations and user context. Built Retrieval-Augmented Generation (RAG) pipeline solving long-term AI memory problem using LangChain orchestration, ChromaDB vector database (1000+ embeddings), semantic search for context retrieval.
- Implemented FastAPI backend for REST APIs and LLM orchestration; Streamlit frontend for conversational UI. Designed 4 slash commands: /remember (save context to memory), /recall (retrieve similar past interactions with <200ms latency), /summarize (generate conversation summaries), /forget (delete specific memories).
- Deployed on Render + Streamlit Cloud ensuring public accessibility. Successfully indexed and retrieved 1000+ conversation vectors with full feature set working flawlessly. Achieved hackathon submission demonstrating production-ready code quality and innovative AI solution.

CareerForge AI — SaaS AI Career Mentorship Platform | In Development

[Github](#)

- Conceptualized production-grade SaaS platform addressing gap: students lack accessible, personalized career mentoring. Integrating Groq API with Llama 3.3 70B for advanced reasoning enabling nuanced career analysis (resume evaluation, skill gap identification, interview strategy, career roadmaps).
- Engineered sophisticated multi-step prompt engineering: resume analysis → skill gap detection → actionable recommendations. Built FastAPI backend supporting 1000+ concurrent users with scalable architecture (authentication, user management, data persistence). Developed Streamlit conversational UI for interactive coaching.
- Actively iterating on features, prompt quality, and UX toward MVP launch. Demonstrates full-stack SaaS capability, product thinking, prompt engineering expertise, and entrepreneurial vision.

EDUCATION

B.Tech – Artificial Intelligence & Data Science | P.A.College of Engineering and Technology | **Current CGPA: 9.17 | 2023–2027 Batch**

ACHIEVEMENTS

- Awards & Hackathons: Pentathlon 2025 Finalist (AICTE) | SIH 2025 Internally Selected | Flipkart GRID 7.0 | Amazon HackOn | Infosys Warrior Badge | GirlScript SoC 2025 | 1st Prize English Poetry Competition | Zoho Cligtrix 2025 Hackathon
- Certifications: Summer Analytics 2025 (IIT Guwahati) | Tata GenAI Data Analytics (Forage) | Deloitte Data Analysis (Forage) | NLP Fundamentals & AI for All (Infosys Springboard) | Quantum Computing Workshop (2024) | AWS AI & ML Scholars Program | AWS Prompt Engineering | Postman API Student Expert